UNIVERSIDAD
NACIONAL
DE COLOMBIA

Prototype 3
Quality Attributes, Part 1

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Software Architecture

2026-I

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1. Objective

The objective of the third delivery of the project is to **redesign the architecture** and generate a third **prototype** of the software system developed as a project.

2. Requirements

2.1. Functional Requirements

- The functional completeness for the software system must be defined by the team.

2.2. Non-Functional Requirements

- The software system must respond to at least four different **security** scenarios:
  - In scenario 1: the software system must implement the **Secure Channel Pattern** as a countermeasure.
  - In scenario 2: the software system must implement the **Reverse Proxy Pattern** as a countermeasure.
  - In scenario 3: the software system must implement the **Network Segmentation Pattern** as a countermeasure.
  - In scenario 4: the software system must implement another pattern defined by the team as a countermeasure.
- The software system must be subject to a **performance** analysis.

3. Delivery

3.1. Artifact

- Team

    - Name (1a, 1b, ..., 2a, 2b, ...)
    - Full names of the team members.
- Software System
    - Name
    - Logo
    - Description
- Architectural Structures
    - Component-and Connector (C&C) Structure
        - C&C View
        - Description of architectural elements and relations.
        - Description of architectural styles and patterns used.
    - Deployment Structure
        - Deployment View
        - Description of architectural elements and relations.
        - Description of architectural patterns used.
    - Layered Structure
        - Layered View
        - Description of architectural elements and relations.
        - Description of architectural patterns used.
    - Decomposition Structure
        - Decomposition View
        - Description of architectural elements and relations.
- Quality Attributes
    - Security
        - Security scenarios.
        - Applied architectural tactics.
        - Applied architectural patterns.
    - Performance and Scalability
        - Performance testing analysis and results.

3.2. Submission Instructions

- The artifact must be created in **.md** format and exported to **.pdf** format.
- The artifact must be named **p3_X.pdf** (where X = the team name [1a, 1b, ..., 2a, 2b, ...]).
- The artifact must be submitted via MiCampus UNAL.
- Only one member of the team must submit the artifact.

3.3. Delivery Deadline

Wednesday, May 20, 2026.

3.4. Delivery Presentation

Thursday, May 21, 2026 (in class).